

CREDIT POLICY RECOMMENDATION REPORT

Digital Lending Portfolio Optimization

Addressed To:

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Digital Lending: Portfolio Optimization Project

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1. Executive Summary

This report presents a data-driven credit policy framework developed through end-to-end analysis of a 10,000-loan synthetic digital lending portfolio. Using exploratory data analysis, predictive modelling (Logistic Regression AUC: 0.699, XGBoost AUC: 0.693), and SHAP-based feature attribution, we have identified the primary drivers of default risk, quantified segment-level profitability, and translated these findings into actionable credit policy recommendations.

Note: This synthetic dataset was designed to stress-test policy frameworks across all risk segments, including high-risk borrowers typically excluded in production portfolios. The 42.5% rate reflects intentional inclusion of Grade E–G borrowers to enable comprehensive policy design. In live production portfolios, default rates typically range from 2–8%; the relative risk differentials and policy recommendations in this report remain directionally valid regardless of the absolute default rate.

The portfolio currently operates at an overall default rate of 42.5%, intentionally elevated to ensure sufficient class representation across all risk segments for robust predictive modelling. Our analysis reveals that this default rate is not uniformly distributed — it is concentrated in specific loan grades, employment segments, and acquisition channels that can be addressed through targeted policy interventions.

Key findings at a glance:

- Grade C delivers the best risk-adjusted return (expected revenue: Rs. 14,154 at 26.8% default rate) and should be the primary growth target.
- Grades E, F, and G account for a disproportionate share of defaults (57%–62%) while offering marginal incremental revenue over safer grades.
- Interest rate and DTI are the top two default predictors (SHAP analysis), suggesting a rate-cap mechanism for borderline borrowers could break the default-rate cycle.
- Salaried borrowers represent the sweet spot: lowest default rate (34.6%), largest volume (2,965), and reliable CLV of Rs. 23,559.
- Agent-acquired customers default at 47.7% vs. 39.2% for Referral — the highest-quality acquisition channel.

Implementation of the recommended policy changes is projected to reduce the portfolio default rate from 42.5% to approximately 26–28%, while preserving 85%+ of revenue-generating volume.

2. Risk Segmentation Findings

2.1 Default Rate by Loan Grade

Loan grade — derived from FICO score and income — is the single strongest structural risk predictor in the portfolio. The gradient from Grade A to Grade G is steep and consistent:

Grade	Default Rate	Avg CLV (Rs.)	Expected Rev (Rs.)	Volume	Risk Tier
A	14.0%	10,523	9,047	328	Low
B	17.6%	13,853	11,414	761	Low

Grade	Default Rate	Avg CLV (Rs.)	Expected Rev (Rs.)	Volume	Risk Tier
C	26.8%	19,343	14,154	1,383	Medium
D	40.6%	23,388	13,902	3,807	High
E	57.0%	29,396	12,629	2,856	Very High
F	60.4%	35,915	14,233	593	Very High
G	62.1%	39,259	14,867	272	Critical

Critical observation: Despite higher gross CLV in Grades F and G (Rs. 35,915 and Rs. 39,259 respectively), their expected revenue after default losses (Rs. 14,233 and Rs. 14,867) is only marginally above Grade C (Rs. 14,154) — yet they carry more than twice the default risk. This is a clear case where chasing higher-rate borrowers destroys value on a risk-adjusted basis.

2.2 Default Rate by Employment Type

Employment type is the second most impactful segmentation variable, with a 20-percentage-point spread between the safest (Salaried) and riskiest (Informal) segments:

Employment Type	Default Rate	Avg CLV (Rs.)	Volume	Risk Assessment
Salaried	34.6%	23,559	2,965	Preferred Segment
Self-Employed	42.6%	24,080	1,445	Conditional Approval
Informal	55.0%	22,911	879	Restricted
Student	53.7%	24,695	460	Restricted

Note on Student segment: Although Students show the highest CLV proxy (Rs. 24,695), this figure is misleading — over half default before completing repayment. The realized revenue from this segment is significantly lower than CLV projections suggest.

2.3 Default Rate by Acquisition Channel

Acquisition channel reflects the quality of borrower screening at the point of onboarding. Agent-driven acquisition has the weakest risk profile, likely due to incentive misalignment where agents prioritize volume over quality.

Channel	Default Rate	Volume Share	Policy Action
Referral	39.2%	25%	Incentivize — offer rate discount
App Organic	41.9%	40%	Maintain — primary growth channel

Channel	Default Rate	Volume Share	Policy Action
Partnership	43.1%	15%	Monitor — quarterly review
Agent	47.7%	20%	Tighten — mandatory verification

2.4 Loan Size, Ticket Size, and Tenure Risk Analysis

Analysis of loan amount buckets within each grade reveals that loan size is a significant risk amplifier for mid-tier grades, but less predictable at extremes:

- Grade A: Very Large loans (>Rs. 150K) actually default less (10.0%) than Large loans (18.5%), suggesting high-income Grade A borrowers requesting large loans are financially robust. No loan cap needed.
- Grade C: Default rate rises consistently from 20.3% (Small) to 37.8% (Very Large). A loan cap at Rs. 75,000 is recommended to contain risk in this segment.
- Grade E: Even small loans default at 45.9%, rising to 66.3% for Very Large. Loan size restrictions alone are insufficient — grade-level restrictions are necessary.

Tenure analysis adds a further dimension to product risk. Counterintuitively, very short tenures carry the highest default risk: 3-month loans default at 50.9% and 6-month loans at 46.1%, compared to 40.3% for 12-month and 38.8% for 36-month loans. This occurs because ultra-short tenures concentrate repayment burden into a narrow window, creating acute cash-flow stress for borrowers. For Grade D–E borrowers specifically, 12-month loans default at 46.9% while 3-month loans reach 55.1% — a nearly 9-percentage-point difference. Shorter tenures reduce total exposure duration but do not reduce per-loan default probability. The recommended tenure policy is therefore: Grade D–E borrowers should be directed to 12–18 month products rather than ultra-short 3–6 month products, balancing exposure duration with repayment manageability.

3. Early Warning Signal Logic

3.1 SHAP-Based Risk Drivers

Our XGBoost model (AUC: 0.693) with SHAP analysis identified the following features as the most impactful default predictors, in order of importance:

Rank	Feature	Business Interpretation	Signal Threshold
1	Interest Rate	High rate = high-risk borrower; rate itself worsens affordability	> 28%
2	DTI	Debt burden relative to income — key liquidity stress indicator	> 0.50
3	Employment Type	Informal/Student borrowers have	Informal or Student

Rank	Feature	Business Interpretation	Signal Threshold
		unstable income streams	
4	Cash Flow Volatility	Irregular income increases repayment risk	> 0.75
5	Acquisition Channel	Agent channel carries adverse selection risk	Agent
6	Credit History	No credit history borrowers show elevated default rates	has_credit_history = 0
7	Installment-to-Income	Monthly installment burden on disposable income	> 0.35
8	Spending Shock	Sudden spending spikes precede defaults	spending_shock = 1

3.2 Rule-Based Early Warning Tiers

Based on the above features, we propose a three-tier early warning classification system:

Tier	Trigger Conditions	Recommended Action	Review Cycle
WATCH	DTI > 0.50 OR spending_shock = 1	Flag for monitoring; send repayment reminder	Monthly
ALERT	DTI > 0.55 AND (installment_to_income > 0.35 OR cash_flow_volatility > 0.75)	Proactive outreach; offer restructuring options	Bi-weekly
CRITICAL	Grade >= E AND DTI > 0.55 AND spending_shock = 1	Immediate intervention; escalate to collections team	Weekly

The rate-trap problem: A key insight from SHAP analysis is that interest_rate is the #1 default predictor. This reflects a self-reinforcing cycle: riskier borrowers receive higher interest rates, which increases their monthly burden (raising DTI and installment-to-income), which in turn increases default probability. Breaking this cycle through tiered rate caps for borderline Grade D borrowers could materially improve outcomes.

4. Credit Policy Recommendations

4.1 Grade-Based Approval Framework

Grade	Decision	Max Loan Amount	Max Tenure	Conditions
A	Approve	No cap	36 months	Standard terms; eligible for loyalty pricing
B	Approve	Rs. 3,00,000	36 months	Standard terms
C	Approve	Rs. 75,000	24 months	Loan cap enforced; rate ceiling at 20%
D	Conditional	Rs. 50,000	18 months	Require income proof; DTI must be < 0.45
E	Restricted	Rs. 25,000	12 months	Salaried only; mandatory credit history
F	Decline	N/A	N/A	Decline unless Salaried with FICO > 640
G	Decline	N/A	N/A	Decline all applications

4.2 Employment-Specific Rules

- Salaried: Full product access across all eligible grades. Priority segment for cross-sell and upsell strategies.
- Self-Employed: Eligible for Grades A–D only. Must submit 3-month bank statement as income proof. DTI cap of 0.45.
- Informal Workers: Eligible for Grades A–C only. Hard loan cap of Rs. 30,000. Require a co-applicant or guarantor for loans above Rs. 15,000.
- Students: BNPL and Education loans only, capped at Rs. 15,000. No personal loans or working capital products.

4.3 Channel-Specific Underwriting Rules

- Agent Channel: Mandatory physical income verification for all loans above Rs. 20,000. Agent performance scorecard review every quarter; deactivate agents with portfolio default rate > 55%.
- Referral Channel: Eligible for a 0.5% interest rate reduction as a trust premium. Referrer accountability: flag if referred borrower defaults within 90 days.
- App Organic: Maintain current underwriting. Primary growth channel — invest in UX improvements to increase conversion of Grade A–C applicants.
- Partnership Channel: Quarterly audit of partner portfolio quality. Renegotiate partnership terms if default rate exceeds 45%.

4.4 Behavioural Guardrails

- DTI Hard Cap: Reject all applications where DTI exceeds 0.65 regardless of grade.
- Installment-to-Income Cap: Monthly installment must not exceed 40% of verified monthly income.
- Spending Shock Flag: Applications flagged with `spending_shock = 1` in the last 30 days require additional manual review before approval.
- Rate-Trap Intervention: For Grade D borrowers with FICO between 665–680, offer a rate 1.5% below the standard Grade D rate in exchange for setting up auto-debit. This reduces default probability while maintaining borrower trust.

4.5 New-to-Credit (NTC) Customer Policy

3,488 customers (34.88%) in our portfolio have no formal credit history. Standard FICO-based scoring excludes them entirely, yet blanket rejection of this segment forfeits a significant growth opportunity — particularly among younger Salaried and Self-Employed borrowers who are creditworthy but underserved by traditional bureaus.

We evaluate NTC customers using three alternative signals that do not require bureau history:

NTC Qualifying Signal	Threshold	Rationale
Installment-to-Income Ratio	Must be below 0.35	Ensures repayment burden is manageable without credit history
Debt-to-Income (DTI)	Must be below 0.50	Caps total debt obligation at safe level for unproven borrowers
Employment Type	Salaried or Self-Employed only	Informal workers and Students excluded due to income instability

NTC customers passing all three checks receive a starter loan capped at Rs. 30,000 with a 12-month tenure. This restricted product limits institutional exposure while giving the borrower an opportunity to establish a repayment track record. After 2 successful on-time repayments, the customer is reclassified into the standard grade framework using observed repayment behavior as a proxy credit signal. This graduation pathway is critical for long-term customer lifetime value — NTC borrowers who convert to standard grades represent a high-loyalty, low-acquisition-cost segment.

5. Projected Impact Assessment

5.1 Default Rate Reduction

Applying the recommended policy framework — primarily restricting Grades F/G, tightening Grade E to Salaried-only, and enforcing the Agent channel rules — is projected to produce the following portfolio changes:

Metric	Current State	Post-Policy (Est.)	Change
Overall Default Rate	42.5%	~26–28%	-14 to -16 pp
Grade F/G Volume	865 loans	~100 loans (Salaried only)	-88% reduction
Agent Channel Default Rate	47.7%	~40% (est.)	-7.7 pp
Grade C Loan Cap Impact	37.8% (Very Large)	Capped at Medium bucket	Risk contained
Portfolio Volume Retained	10,000 loans	~7,500–8,000 loans	-20 to -25%
Expected Revenue (Avg/Loan)	Rs. 13,200 (est.)	Rs. 13,600 (est.)	+3% improvement

The volume reduction of 20–25% is concentrated in the lowest expected-revenue segments (Grades F/G contribute only ~8% of total portfolio volume but disproportionately inflate the default rate). The net effect is a cleaner, more profitable portfolio.

5.2 Revenue Quality Improvement

Grade C is the optimal growth target: it offers the highest expected revenue per loan (Rs. 14,154) with a manageable default rate. Shifting 20% of current Grade D volume into Grade C-quality borrowers (through stricter Grade D underwriting that upgrades marginal applicants) could add Rs. 2–3 Cr in incremental annual expected revenue at portfolio scale.

The Salaried segment (2,965 customers, 34.6% default rate) represents the core profitable base. Increasing the share of Salaried borrowers from the current 45% to 55% of total approvals — achievable through the channel and employment policy changes above — would reduce portfolio default rate by an estimated 4–5 percentage points independently.

6. Leadership Monitoring Dashboard

Senior leadership should track the following metrics on a monthly basis to proactively manage portfolio risk and growth:

Metric	Current Baseline	Target Threshold	Alert Trigger
Portfolio Default Rate	42.5%	< 30%	> 35%
Grade A-C Share of Approvals	~25%	> 40%	< 30%
Agent Channel Default Rate	47.7%	< 42%	> 50%
DTI > 0.5 in New Approvals	~35% of portfolio	< 20%	> 30%

Metric	Current Baseline	Target Threshold	Alert Trigger
Early Warning — WATCH Tier	Not tracked	Establish baseline	Month-on-month increase
30-Day Delinquency Rate	Not tracked	< 15%	> 20%
Referral Channel Share	25%	> 30%	< 20%
Salaried Borrower Share	~45%	> 55%	< 40%

7. Conclusion

The digital lending portfolio faces a structural default challenge that, while significant, is tractable through data-driven policy design. Our analysis demonstrates that the 42.5% default rate is not an industry inevitability but a consequence of specific, identifiable patterns in borrower grade distribution, employment mix, and acquisition channel quality.

The three most impactful levers available to the institution are:

- Grade discipline: Concentrating approvals in Grades A–C (where expected revenue is highest on a risk-adjusted basis) and strictly restricting Grades F–G.
- Employment prioritization: Actively growing the Salaried segment through targeted product design and channel incentives.
- Channel quality improvement: Reforming Agent channel underwriting and incentivizing Referral channel growth.

Together, these measures are projected to reduce the portfolio default rate by 14–16 percentage points while retaining 75–80% of current loan volume and improving per-loan expected revenue. The early warning signal framework provides a continuous monitoring mechanism to ensure policy effectiveness is tracked and refined over time.

We recommend immediate implementation of the Grade F/G restrictions and Agent channel reforms as Phase 1, with the behavioural guardrails and rate-trap intervention framework as Phase 2 over the following quarter.

Project Team

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